

Low Fouling Polysulfobetaines with Variable Hydrophobic Content

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Amphiphilic polymer coatings combining hydrophilic elements, in particular zwitterionic groups, and hydrophobic elements comprise a promising strategy to decrease biofouling. However, the influence of the content of the hydrophobic component in zwitterionic coatings on the interfacial molecular reorganization dynamics and the anti-fouling performance is not well understood. Therefore, coatings of amphiphilic copolymers of sulfobetaine methacrylate 3-[N-2'-(methacryloyloxy)ethyl-N,N-dimethyl]-ammonio propane-1-sulfonate (SPE) are prepared which contain increasing amounts of hydrophobic *n*-butyl methacrylate (BMA). Their fouling resistance is compared to that of their homopolymers PSPE and PBMA. The photo-crosslinked coatings form hydrogel films with a hydrophilic surface. Fouling by the proteins fibrinogen and lysozyme as well as by the diatom *Navicula perminuta* and the green algae *Ulva linza* is assessed in laboratory assays. While biofouling is strongly reduced by all zwitterionic coatings, the best fouling resistance is obtained for the amphiphilic copolymers. Also in preliminary field tests, the anti-fouling performance of the amphiphilic copolymer films is superior to that of both homopolymers. When the coatings are exposed to a marine environment, the reduced susceptibility to silt incorporation, in particular compared to the most hydrophilic polyzwitterion PSPE, likely contributes to the improved fouling resistance.

animals, on man-made surfaces that are exposed to aqueous media. The accumulation of these organisms impacts most severely the surface and interface properties of objects exposed to the sea, which thus lose their original function and performance. For instance, ships, marine energy turbines, harbor installations, and oceanographic sensors are prone to marine biofouling leading to high maintenance costs.^[1] Ships, for example, experience higher shear forces due to rougher surfaces resulting from biofouling, increasing fuel consumption, and imposing extra costs for hull cleaning.^[2] The current strategy to control biofouling is the use of coatings releasing biocides such as copper compounds,^[3] which are increasingly restricted, thus creating the demand for non-toxic alternatives.^[4,5] An ideal coating combines an inhibiting effect that reduces initial settlement of foulers, and/or facilitates an easy release of attached organisms under low shear stress (fouling-release).^[6]

Many of the hydrophilic coating materials explored for antifouling use have been chosen due to their performance in the

biomedical field, where they were known to improve hemocompatibility, and/or to reduce toxicity, inflammatory, or immune responses. These investigations have mostly focused on non-ionic hydrophilic polymers such as polyethyleneglycol (PEG),

1. Introduction

The term biofouling denotes the unwanted accumulation of biological matter, including proteins, (micro)organisms, plants, and

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polysarcosine, or more recently poly(2-hydroxypropylmethacrylamide) in order to minimize, or even suppress biofouling.^[7,8] Alternatively, hydrophilic zwitterionic surfaces have also demonstrated high resistance to biofouling and high biocompatibility for both technical and biomedical uses.^[9–13] Zwitterions are charge-neutral molecules that contain pairs of positive and negative charges, which allow electrostatic interactions with surrounding water molecules.^[14] As a result, a stable hydration shell is formed, which is assumed to counteract the adsorption and binding of proteins and the settlement of micro- and macrofoulers,^[9,10,14,15] because the bond formation between the organisms' adhesives and the surface is inhibited.^[16] However, the performance of such polymeric coatings in the field is often limited due to a lack of resilience.^[9,17,18] While the chemical stability seems to be the lesser problem when appropriate polymers are employed, for example, such having a polymethacrylate backbone,^[18,19] the mechanical strength is generally low due to their hydrogel character. Based on many laboratory model studies, very hydrophilic and highly swollen zwitterionic coatings confer high fouling resistance, but the extensive swelling tends to impair the mechanical properties.

Besides hydrophilic polymers, amphiphilic polymers containing adjacent hydrophilic and hydrophobic moieties have emerged as promising candidates for marine antifouling coatings.^[4,20–23] Despite the low surface energy, hydrophobic polymers, in general, show little resistance against non-specific adhesion,^[24] but their combination with hydrophilic compounds can lead to inert surfaces with superior fouling-release properties.^[25] For amphiphilic systems, not only the averaged content of hydrophilic and hydrophobic moieties is relevant, but also the distance between the two different groups which can organize into nanodomains.^[26] Coatings with a compositional control over hydrophilic and hydrophobic components were shown to exhibit resistance against a broader range of species, which was assigned to the formation of microdomains of different properties on the surface of the polymer films.^[21,22,27] Such amphiphilic systems can be conveniently prepared by copolymerization of hydrophilic and hydrophobic monomers. Seemingly, compositional and morphological heterogeneity at the nanoscale is crucial for a successful amphiphilic low-fouling coating, while microphase segregation should be avoided.^[28]

Possibly beneficial properties of zwitterionic amphiphilic coatings for fighting marine biofouling have occasionally been explored. For instance, the incorporation of 1–20 mol% of the sulfobetaine 3-[N-2'-(methacryloyloxy)ethyl-N,N-dimethyl]ammonio propane-1-sulfonate (SPE) into a hydrophobic lauryl methacrylate-based coating increased the anti-fouling resistance slightly.^[29] The increase in surface roughness was, however, too great to form effective coatings, most likely due to phase separation during the preparation of the films. Also, low adhesion of proteins was reported for mixed amphiphilic surfaces containing carboxybetaine methacrylates and hydrophobic *n*-butyl methacrylate (BMA).^[30] The incorporation of 5–15 wt% of carboxybetaine methacrylate into a hydrophobic ethylene glycol dicyclopentenyl ether acrylate matrix improved the anti-fouling efficiency significantly.^[23] The addition of zwitterionic additives including carboxybetaine methacrylate to hydrophobic ethylene glycol dicyclopentenyl ether methacrylates and the analogue acrylates showed that the choice of the hydrophobic component is

important for anti-fouling behavior. The added carboxybetaine methacrylate outperformed its acrylic analogue in the field.^[31] Recently, copolymers of BMA with various ionic comonomers were prepared and characterized, among them poly(BMA-co-SPE) at a molar ratio of 9:1.^[32] Relatively high amounts of non-freezable water strongly bound to the polymer were found for these zwitterionic amphiphilic copolymers. Such coatings showed significantly lower barnacle settlement and a high release of fouling organisms at relatively low shear forces compared to pure BMA. The authors speculated that the swelling of the polymers in seawater might contribute to the high fouling-release efficacy. Note in this context, that the fouling resistance of SPE-based coatings was also improved with increasing amounts of the hydrophobic photo-crosslinker 2-(4'-benzoylphenoxy)ethyl methacrylate (BPMEA).^[33] It is not clear yet, to what extent the improvement derives from the higher crosslink density, which seems highly effective for, for example, reducing the protein fouling of nano carriers,^[8] or to the increasingly amphiphilic character of these polymers. Also, statistical copolymers of BMA containing up to 30 mol% of SPE on poly(methylmethacrylate) (PMMA) were reported to reduce the adhesion of macrophages and fibroblast cells as well as some pathogenic bacteria compared to uncoated PMMA, the effects increasing with the sulfobetaine content.^[34] Analogously in the biomedical context, zwitterionic copolymers of BMA with a phosphobetaine monomer, namely phosphatidylcholine methacrylate (MPC), have been used successfully for more than two decades for coating medical devices and implants, excelling in their high hemocompatibility and resistance against the adhesion of proteins, bacteria, and cells.^[35] Although the success of the latter copolymers is often credited to the unique structure of the building block MPC and its particular interaction with water molecules,^[36] the amphiphilic character of these copolymers may also play an important role in their performance. Note that also copolymers of BMA with sulfobetaine comonomers (instead of MPC) seem to exhibit comparably advantageous polymer–water interactions.^[12] In contrast, a recent study on coatings of copolymers of BMA with the analogous carboxybetaine comonomer reported low adhesion of fibroblast cells but strong fouling by proteins such as fibrinogen, which was attributed to an overall positive ζ -potential.^[13]

Such studies demonstrated improved fouling resistance of hydrophobic polymer coatings when rendering them partially hydrophilic. Still, in reverse, for strongly hydrophilic polymers and hydrogels, the general perception is that the fouling resistance is more or less compromised when introducing additional hydrophobic building blocks, not only in the biomedical field.^[37] Nevertheless, considering these observations, amphiphilic copolymers of SPE with BMA seem to be attractive candidates for marine fouling-resistant coatings. While the phosphobetaine moiety in MPC-based polymers is rather sensitive to hydrolytic degradation in marine environments, sulfobetaines (as in SPE) are much more resilient.^[38] Still, up to now, a systematic study of the anti-fouling potential of copolymers of SPE and BMA, in particular with high zwitterion contents, has been missing. In this study, we prepared amphiphilic coatings based on the sulfobetaine monomer SPE, which contain 10, 30, and 50 mol% of hydrophobic BMA (see **Figure 1**). All copolymers also contain 1 mol% of the photo cross-linker BPMEA, which assured the stability of the coatings in aqueous media after a photo-curing

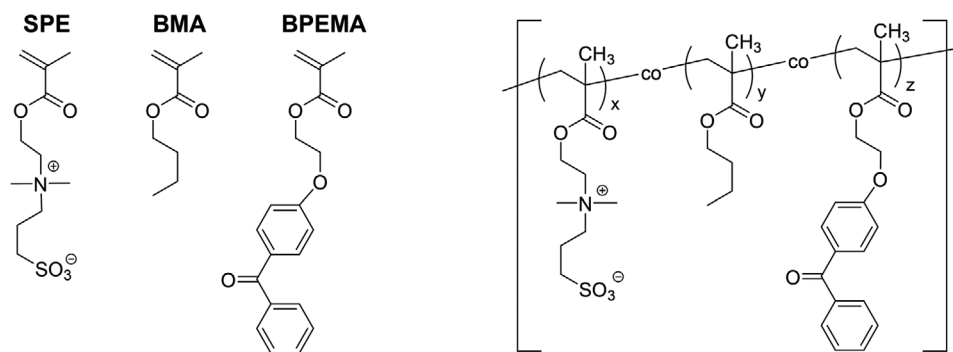


Figure 1. Chemical structure and abbreviations of the monomers used (left) and the copolymers produced (right). For the photo-reactive pseudo-homopolymers PSPE ($x = 0.99$, $y = 0$) and PBMA ($x = 0$, $y = 0.99$) as well as the terpolymers coP-SPE-9-BMA-1 ($x = 0.89$, $y = 0.10$), coP-SPE-7-BMA-3 ($x = 0.69$, $y = 0.30$), and coP-SPE-5-BMA-5 ($x = 0.495$, $y = 0.495$), the content of BPEMA is $z = 0.01$.

Table 1. Polymer composition, molar masses, and sample codes.

Sample	BMA content [mol%]	SPE content [mol%]	BPEMA content [mol%]	Yield [%]	M_w^{app} [kg mol ⁻¹] ^{a)}	$\bar{D}^{b)}$
PBMA ^{c)}	99 (99) ^{e)}	–	1 ± 0.3	77	140	2.8
coP-SPE-5-BMA-5	46 ^{d)} (49.5) ^{e)}	53 ^{d)} (49.5) ^{e)}	1 ± 0.3	84	240	3.1
coP-SPE-7-BMA-3	27 ^{d)} (30) ^{e)}	72 ^{d)} (69) ^{e)}	1 ± 0.3	73	320	3.3
coP-SPE-9-BMA-1	11 ^{d)} (10) ^{e)}	88 ^{d)} (89) ^{e)}	1 ± 0.3	63	420	4.7
PSPE ^{c)}	–	99 (99) ^{e)}	1 ± 0.3	72	490	5.4

^{a)} Apparent weight average molar mass by SEC, eluent 50 mM CF₃COONa in HFIP, calibration by PMMA standards; ^{b)} Apparent dispersity (M_w^{app}/M_n^{app}) by SEC, eluent 50 mM CF₃COONa in HFIP, calibration by PMMA standards; ^{c)} Data from ref. [9]; ^{d)} Values are precise within ± 10 rel% (for details see Supporting Information); ^{e)} Values put in parentheses indicate the comonomer content in the original reaction feed (see Supporting Information).

step.^[9,18] Identically photo-cured coatings of the underlying photo-reactive “homopolymers” PSPE and PBMA served as references. The fouling resistance was evaluated in laboratory assays using proteins and the model fouling organisms *Navicula perminuta* (*N. perminuta*) and *Ulva linza* (*U. linza*). The swelling of the coatings and the surface rearrangement of the hydrophilic and hydrophobic molecular fragments were monitored with captive bubble contact angle (CBCA) measurements. To assess a potential use in the marine environment, the inertness against sediment was tested additionally by a short-term field experiment.

2. Results

2.1. Design and Synthesis of the Amphiphilic Copolymers

The amphiphilic SPE/BMA copolymers were obtained by free radical copolymerization of the zwitterionic methacrylate monomer SPE in trifluoroethanol (TFE) with varying amounts of hydrophobic butyl methacrylate BMA and a fixed amount of the benzophenone-bearing methacrylate BPEMA (Figure 1) as photo cross-linker. Detailed procedures for the synthesis and analytical data are provided in the Supporting Information. The content of the photo cross-linker BPEMA was kept constant at 1 mol% in the reaction mixture, so that the crosslinking density in all photo-cured polymer coatings should be similar. The content of the hydrophobic comonomer BMA in the reaction mixture was increased from 10 to 30, and eventually to 50 mol% in

the copolymer series. The compositional analysis of the ternary copolymers, which are all hygroscopic, was not trivial. The combined data from elemental analysis and the integration of constitutional repeat unit (CRU)-specific signals in the 1 H NMR spectra revealed that the contents of BMA and BPEMA in the zwitterionic copolymers corresponded to the composition of the copolymerization mixtures within the precision of the data, as summarized together with other key molecular parameters in Table 1.

As the monomer conversions were high, the analytical data cannot rigorously prove, that all copolymerization reactivity ratios are virtually at unity, that is, that the system undergoes truly ideal azeotropic copolymerization. Only in this case, the copolymers formed at different reaction times have a uniform composition, and the distribution of the CRUs within a given chain follows random statistics. To achieve this, comonomers were selected which bear the identical polymerizable moiety, namely methacrylate, and the copolymerization was conducted under homogeneous conditions in a good solvent for all components, namely TFE, as all these measures a priori favor ideal azeotropic copolymerization. Nevertheless, the large differences in monomer polarities must be kept in mind. Studies of copolymerization of SPE and BMA in well-compatible solvents including ionic liquids revealed deviations from ideal behavior when kept at low conversion rates, indicating a tendency for micro blockiness in the sequence of zwitterionic and hydrophobic monomers within the polymer chains.^[39] High molar mass copolymers were synthesized (Table 1), with broad molar mass distributions as is typical for free radical polymerizations carried to high conversions. Accordingly,

Table 2. Wettability of the coatings by water. Static water contact angles (WCA) were determined on the dry polymers in air, the captive bubble water contact angles (CBCA) were determined after immersion in salt water. The contact angle decrease ($|\Delta CA|$) is the difference between WCA and CBCA. Reported errors are the standard errors after 20 measurements (WCA, dry) and 5 measurements after 7 days immersion time (CBCA, underwater). The APTMS-pretreated support is included as reference substrate prior to polymer coating.

	WCA [°]	CBCA [°]	$ \Delta CA $ [°]
PBMA	73 ± 2	42 ± 2	31
coP-SPE-5-BMA-5	70 ± 1	14 ± 2	56
coP-SPE-7-BMA-3	85 ± 2	13 ± 1	72
coP-SPE-9-BMA-1	60 ± 1	10 ± 1	50
PSPE	25 ± 3	10 ± 1	15
APTMS	56 ± 4	–	–

most copolymer chains contain several photo-crosslinker groups as is needed to prepare stable films with a low leachable content after curing, despite the average content of only about 1 mol% of crosslinker. Thermogravimetric analysis of the copolymers showed the onset of degradation at about 240 °C, as well as 5–10% of weight loss around 100 °C indicative of ≈ 1 –2 molecules of water tightly bound to the sulfobetaine moiety. Differential scanning calorimetry run up to 200 °C did not indicate any glass transition. Upon increasing incorporation of the hydrophobic BMA units into the copolymers, the solubility of the copolymers was somewhat altered compared to the solubility of the homopolymer PSPE. While all copolymers were still soluble in TFE and HFIP, they did not dissolve anymore in hot water, but only in brine, and this only if the BMA content did not exceed 30 mol%. Simple alcohols such as methanol or ethanol, chloroform, mixtures of the latter, and all aprotic solvents tested (including DMF, NMP, DMSO, acetonitrile, acetone, ethyl acetate, THF, CH_2Cl_2 , pyridine, or benzene) remain non-solvents, as noted before for other hydrophobically modified polysulfobetaines with a substantial content of betaine groups.^[40]

2.2. Preparation of Thin Film Polymer Coatings and Surface Characterization

The polymers were applied as thin coatings to surfaces by spin coating their solutions in TFE following previously published protocols.^[9,18] Thin, homogeneous, and smooth coatings with reproducible properties were obtained. The coatings were stabilized by photo-curing, exploiting the photo-reactivity of the benzophenone moiety in the comonomer BPMA. This enabled simultaneous cross-linking of the polymer chains within the coatings as well as their covalent anchoring to surfaces containing aliphatic CH moieties via the so-called C,H-insertion crosslinking mechanism.^[9,41] The wettability of the coatings was characterized by water contact angle (WCA) goniometry (Table 2). The coating of the pure polyzwitterion PSPE was the most hydrophilic one, exhibiting a dry contact angle of $25 \pm 3^\circ$. For the copolymer coatings, the wettability was lower resulting in contact angles of $60 \pm 1^\circ$ for coP-SPE-9-BMA-1, $85 \pm 2^\circ$ for coP-SPE-7-BMA-3, and $70 \pm 1^\circ$ for coP-SPE-5-BMA-5. PBMA produced hydrophobic coatings with a WCA of $73 \pm 2^\circ$. Interestingly, a content of

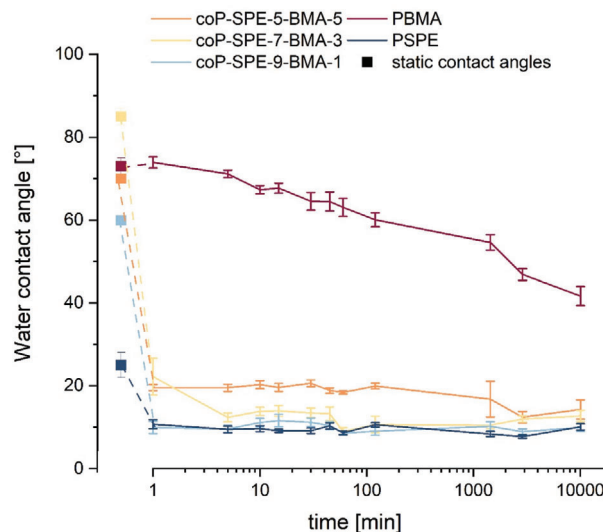


Figure 2. Water contact angle of the coatings. Captive bubble contact angle (CBCA) goniometry of air bubbles on the polymer coatings in saltwater over 7 days of immersion. Error bars indicate the standard error obtained from 5 air bubbles on the same slide. For comparison, squares indicate the static water contact angles in air with the standard error after 20 measurements on different slides. Full and dashed lines are meant as guide to the eye.

only 10 mol% of hydrophobic BMA in the copolymers was sufficient to raise the WCA values substantially compared to the pure PSPE coatings. We noted also that according to expectations, WCA values increased with increasing content of hydrophobic comonomer BMA. To our surprise, the somewhat higher WCAs found for coP-SPE-7-BMA-3 compared to pure PBMA and coP-SPE-5-BMA-5 deviated from this relation.

In order to understand the extent of polymer surfaces rearrangement in aqueous environments, CBCA were measured under water. PBMA had a contact angle of 70° immediately after the immersion in saltwater, which decreased slowly stabilizing at around 42° (Figure 2). This evolution of CBCA was most likely caused by a slow rearrangement of the polymer to expose preferentially the more hydrophilic carboxyl groups toward the polymer–artificial seawater (ASW) interface, thereby burying the hydrophobic butyl side chains inside the polymer film. In contrast, the most hydrophilic polyzwitterions, namely PSPE and coP-SPE-9-BMA-1, exposed hydrophilic surfaces immediately after immersion with a CBCA of around 10° through the whole immersion period. The copolymers coP-SPE-BMA with higher BMA contents showed intermediate behavior. Sample coP-SPE-7-BMA-3 exhibited an initial CBCA of 22° , which decreased within 5 min to 13° . Sample coP-SPE-5-BMA-5 with a further increased BMA content had a WCA of 70° and rearranged within the first minute of immersion in ASW to form a rather hydrophilic surface with a CBCA of 14° .

The analysis of the contact angle changes (Table 2) showed that the WCAs strongly decreased upon immersion in saltwater (CBCA). While for samples with lower SPE content, the change in contact angle increased with an increasing amount of incorporated SPE, the change decreased again after reaching its maximum of 72° for coP-SPE-7-BMA-3. PBMA without SPE units can

only rearrange by moving the more polar ester group close to the backbone to the outside. This conformational rearrangement had only a limited effect, resulting in a decrease of the contact angle by 31°. Sample coP-SPE-5-BMA-5 showed a more pronounced decrease of the contact angle by 56° upon immersion in saltwater due to the presence of the SPE side chains. These allowed for rearrangements that led to an exposure of the hydrophilic SPE groups toward the aqueous phase, resulting in much lower final contact angles (14° compared to 42° for PBMA). This trend continued with coP-SPE-7-BMA-3 featuring a Δ CA of 72°. By increasing the SPE content further in coP-SPE-9-BMA-1, the change in contact angle was less prominent. The polymer films showed a much more hydrophilic surface in their dry state, indicating that many SPE groups were already exposed at the interface. This limited the possibilities to further decrease the contact angle by segmental rearrangements when immersed in water. Nevertheless, even for the polyelectrolyte PSPE that lacked hydrophobic side chains, a moderate, but notable reduction in contact angle by 15° was observed when exposed to saltwater.

To ensure that the coatings were stable during the biological measurements, all samples were immersed in saltwater, which contained all essential salts but no organic components.^[42] There was no substantial decrease in the thickness of the dry coating during 7 days of immersion of the copolymers. These findings obtained by spectroscopic ellipsometry were supported by the CBCA measurements as there was no significant change in the contact angle after the initial rearrangement discussed above. As all copolymer coatings retained their thickness and contact angle over time, they were suited for biological testing without any risk of delamination or disintegration during the duration of the tests.

The water uptake of the polymer coatings in an atmosphere saturated with water vapor was studied by spectroscopic ellipsometry. Such a measurement characterizes the water affinity of the coatings.^[43,44] PBMA did not show any significant swelling due to its hydrophobic nature, which made water uptake unfavorable (Figure 3). Therefore, it was not possible to determine the rate constant of water uptake. PSPE films increased their thickness to 134% within the first 30 min, without reaching full saturation. All mixed amphiphilic coatings swelled less than PSPE, reaching relative thicknesses of 127% for coP-SPE-7-BMA-3, 124% for coP-SPE-9-BMA-1, and 113% for coP-SPE-5-BMA-5. The swelling was close to its maximum after the first 10 min of exposure to water-saturated air. The fitted kinetic constant k_w was between 0.92 for coP-SPE-9-BMA-1 and 1.63 for coP-SPE-7-BMA-3 (Table 3) without a clear trend regarding the composition of the polymers. The exponent p was fitted globally due to the assumption that the degree of diffusion limitation^[45] was equal for all polymers, giving $p = 0.60 \pm 0.01$.

2.3. Protein Adsorption

The resistance of amphiphilic copolymers coP-SPE-BMA against non-specific adsorption of proteins was evaluated by surface plasmon resonance spectroscopy (SPR, Figure 4). The shift in plasmon resonance before and after the injection of fibrinogen and lysozyme provides insight into the amount of irreversibly bound

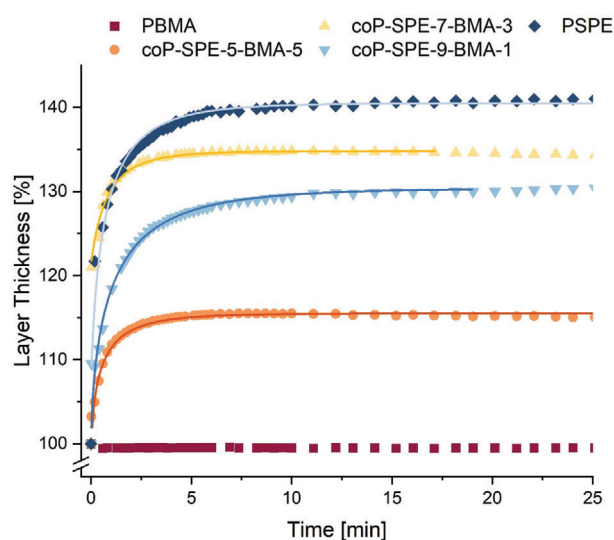


Figure 3. Exemplary time-dependent relative swelling curves of the polymer coatings in a saturated water atmosphere measured by spectroscopic ellipsometry (symbols). The lines represent the fit of the swelling using Equation (1).

Table 3. Uptake of water from a water-saturated atmosphere measured by spectroscopic ellipsometry. Kinetic constants were derived assuming a diffusion limited pseudo first-order kinetics. The reported relative increases are the average of three measurements on three replicate slides. p was fitted globally for all curves resulting in $p = 0.60 \pm 0.01$. PBMA did not show enough swelling to fit a rate constant.

Polymer	% increase of the layer thickness	k_w [min ⁻¹]	t_0	R^2
PBMA	0.1 ± 0.5	—	—	—
coP-SPE-5-BMA-5	13 ± 1	1.353 ± 0.004	0.039 ± 0.002	0.996
coP-SPE-7-BMA-3	27 ± 6	1.637 ± 0.009	-0.410 ± 0.008	0.926
coP-SPE-9-BMA-1	24 ± 5	0.922 ± 0.002	0.007 ± 0.002	0.999
PSPE	34 ± 6	1.290 ± 0.004	-0.041 ± 0.003	0.998

protein. The two model proteins used differ in size and net-charge: fibrinogen has a slightly negative charge at pH 7.4 (pI 5.5, molar mass 340 kDa),^[46] whereas lysozyme has a distinct positive charge (pI 10.9, molar mass 14 kDa).^[47] For fibrinogen, the adsorption was highest on the hydrophobic reference PBMA, while coP-SPE-5-BMA-5 exhibited a more than 200-fold lower protein adsorption. The protein adlayers on the polymers PSPE, coP-SPE-7-BMA-3, and coP-SPE-9-BMA-1 were negligible and too low to allow discrimination between these surfaces at a level of significance of 5%.

Compared to fibrinogen, the adsorption of lysozyme on the surfaces was slightly more differentiated. While PBMA was again contaminated to the largest extent, PSPE exhibited a significantly higher protein resistance (Figure 4, $p < 0.05$, post-hoc Tukey test). The amphiphilic copolymers showed the highest resistance against lysozyme. The coating of coP-SPE-7-BMA-3 had no detectable protein adsorption layer, while the adlayers on coP-SPE-5-BMA-5 and coP-SPE-9-BMA-1 were already very small.

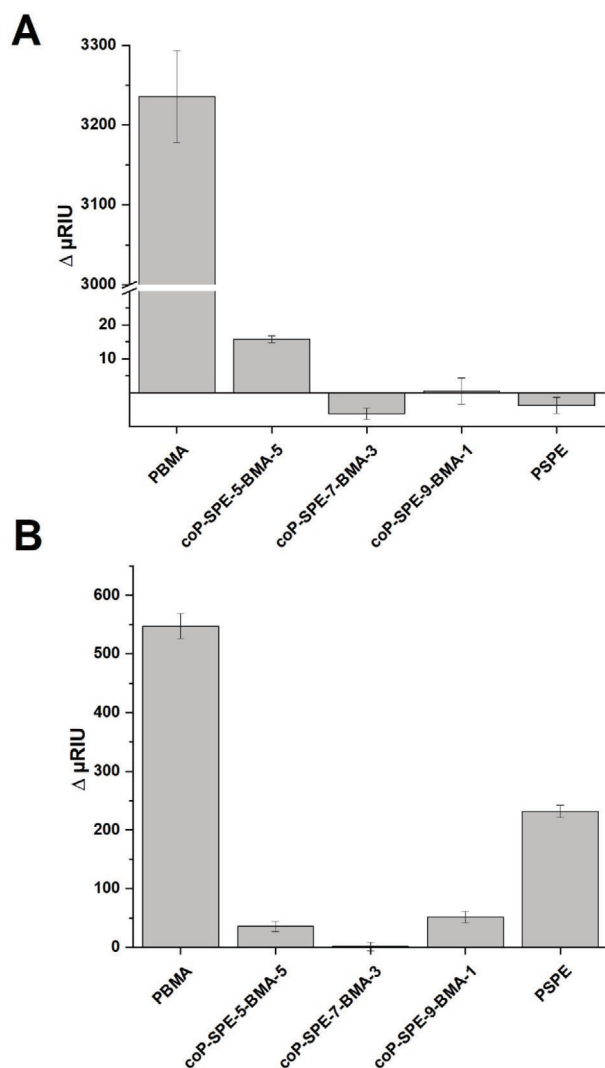


Figure 4. Irreversible adsorption of A) fibrinogen and B) lysozyme on the SPE-BMA copolymers determined by SPR at a flow rate of $10 \mu\text{L min}^{-1}$. Error bars indicate the standard error ($n = 3$).

Remarkably, the transient adsorption during the injection phase of the proteins was not particularly low, but the subsequent buffer injection at a flow speed of $10 \mu\text{L min}^{-1}$ removed the lysozyme nearly completely from the coated surfaces (Figure S1, Supporting Information). Thus, the amount of irreversibly bound protein was much lower than for pure PBMA and even pure PSPE coatings. This finding compares well with a recent report on micellar nanoparticles of copolymers of BMA and SPE, for which the statistical copolymers bound proteins and hydrophobic compounds much weaker than both the PBMA core and the PSPE shell of micellar nanoparticles of the analogous block copolymers.^[36] The extent of protein adsorption was not statistically significantly different for the various copolymers coP-SPE-5-BMA-5, coP-SPE-7-BMA-3, and coP-SPE-9-BMA-1 ($p > 0.05$). However, the differences between the copolymers and the pure PSPE and PBMA polymers were statistically significant ($p < 0.05$).

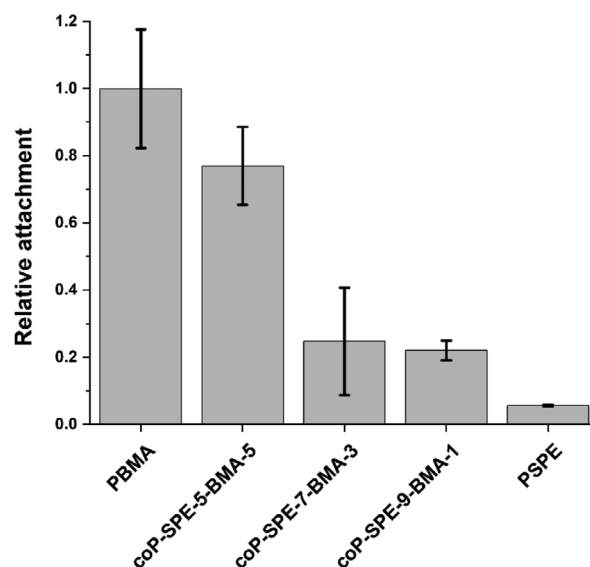


Figure 5. Dynamic microfluidic diatom attachment assay on the polymer coatings, normalized to the attachment on PBMA, at a wall shear stress of 0.16 Pa for 90 min . Error bars indicate the standard error for 3 measurements on 3 replicates with 30 fields of view each.

2.4. Marine Diatom Accumulation

The marine diatom *N. perminuta* is a frequently occurring fouling organism in the marine environment and an established model organism for biofouling studies.^[48,49] The diatom is non-motile and when exposed to surfaces under dynamic conditions, the relative diatom densities provide an estimate of the fouling release properties of the surface.^[48] Figure 5 summarizes the relative attachment of *N. perminuta* after a 90-min dynamic microfluidic accumulation assay at a wall shear stress of 0.16 Pa . All data were normalized to PBMA, which had the highest density of attached diatoms. The increasing incorporation of zwitterionic SPE units into the coatings reduced the accumulation of the diatoms, with PSPE showing the lowest number of attached cells (Figure 5). The mixed copolymers showed a significant ($p < 0.05$) decrease of the relative diatom density from coP-SPE-5-BMA-5 to coP-SPE-7-BMA-3 and coP-SPE-9-BMA-1, but the small difference between the latter two copolymers was not statistically significant ($p > 0.05$).

2.5. *U. linza* Zoospore Settlement and Removal

In contrast to marine diatoms, zoospores of the green algae *U. linza* actively explore surfaces to settle on a favorable spot.^[50,51] Figure 6 summarizes the results of settlement and removal experiments with *U. linza* on the series of amphiphilic copolymers. Settlement was highest on the hydrophobic PBMA but was significantly reduced on the hydrophilic PSPE. All mixed amphiphilic polymers had a significantly lower settlement compared to both homopolymers ($p < 0.05$, post-hoc Tukey test). Among the copolymer coatings, the settlement was highest on coP-SPE-9-BMA-1, while settlement was lowest and not significantly different ($p > 0.05$) on copolymers coP-SPE-7-BMA-3 and coP-SPE-5-BMA-5.

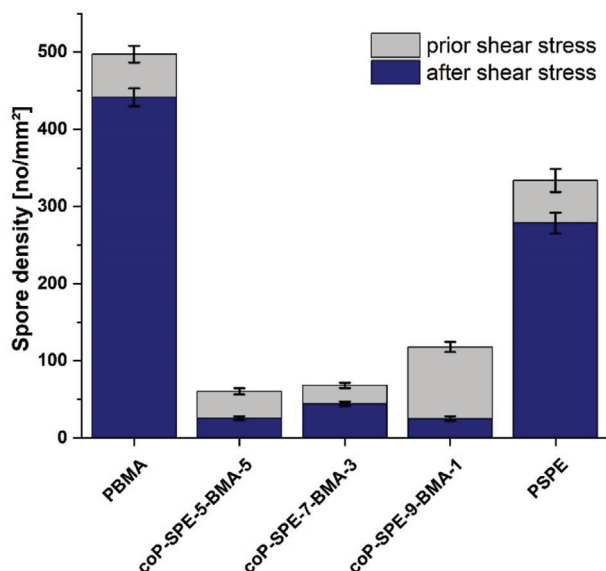


Figure 6. Density of settled *U. linza* zoospores after a 45-min settlement period (grey) and after exposure to a turbulent water flow exerting a wall shear stress of 52 Pa (blue). Error bars indicate the standard error, $n = 90$.

The attachment strength of settled spores on the coatings was challenged by exposure to a wall shear stress of 52 Pa created by a turbulent water flow. A high removal indicates a low attachment strength of the spores on the surfaces. As observed for the direct settlement, the settlement strength was lower on the amphiphilic polymers than on the parent homopolymers, that is, 5% and 8% of the attached spores were removed from the hydrophilic PSPE and the hydrophobic PBMA, respectively. The removal of the spores from the amphiphilic coatings amounted to 38% for coP-SPE-5-BMA-5, 21% for coP-SPE-7-BMA-3, and 76% for coP-SPE-9-BMA-1. After the application of a turbulent flow, the pure PSPE and PBMA coatings still had the highest spore density while all copolymer coatings had a significantly lower number of adhering spores ($p < 0.05$). Among the amphiphilic copolymer coatings, the differences after the fouling removal step were not statistically significant ($p > 0.05$). Altogether, all copolymers show similarly high fouling resistance, but apparently, the more hydrophobic copolymers provide a somewhat higher antifouling effect, while the more hydrophilic copolymers show a somewhat more effective fouling release.

2.6. Sediment Incorporation

Previously, we found that hydrophilic polymer coatings are prone to the incorporation of mineral particles when tested in the field, which reduced the anti-fouling performance dramatically.^[17] We designed a laboratory model to challenge the coatings by incubation in an autoclaved sediment suspension for 2 h before the amount of incorporation of particles was determined photometrically.^[17] The amount of incorporated sediment rose with increasing hydrophilicity of the copolymers and swelling of the coatings in a non-linear fashion. For a BMA incorporation level of 10 mol%, a 38% decrease in sediment incorporation was achieved, while a further increasing BMA content approached a

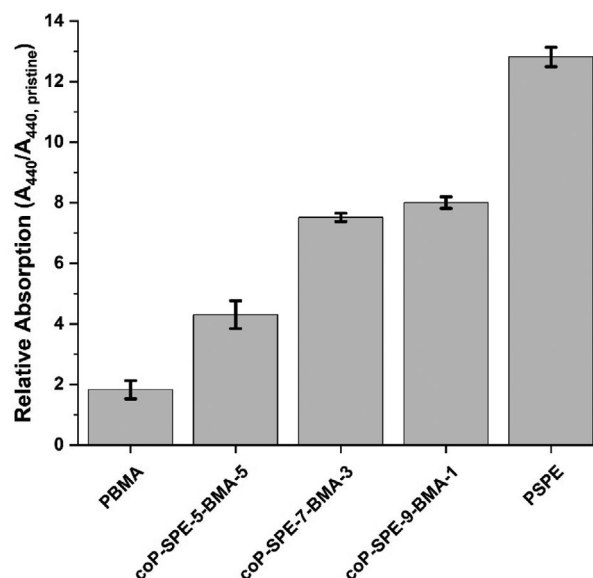


Figure 7. Silt uptake of the coatings determined as an increase of the relative absorbance of the coatings after 2 h sediment immersion. The absorbance of pristine coating and after incubation in sediment was measured at a wavelength of 440 nm. Error bars indicate the standard error after six measurements on two replicate slides.

linear relationship between BMA content and sediment incorporation (Figure 7). The decreasing sediment incorporation with an increasing BMA content was statistically significant for all compositions ($p < 0.05$) except for the direct comparison of coP-SPE-9-BMA-1 and coP-SPE-7-BMA-3 ($p > 0.05$).

2.7. Dynamic Short-Term Field Exposure

To verify the trends observed in the laboratory assays in the presence of a complex fouling community in a real ocean environment, experiments were carried out using coatings attached to a rotating disk setup at the field test site located in Port Canaveral, Florida. After the 7-day immersion period, the biofilm coverage was determined via the autofluorescence of chlorophyll by fluorescence microscopy. In general, the biofilm coverage was significantly higher on coatings with high SPE content (Figure 8). There was no statistical difference between the weak performances of PSPE and coP-SPE-9-BMA-1 with 31% coverage each ($p > 0.05$). Thus, the incorporation of 10 mol% of BMA did not yet noticeably alter the coating performance in the field. Upon the incorporation of 30% BMA, copolymer coP-SPE-7-BMA-3 had a 4 % lower coverage, indicating a slight but statistically significant improvement ($p < 0.05$). A major improvement of the anti-fouling performance was observed for copolymer coP-SPE-5-BMA-5 with the highest BMA content, which reduced the coverage by a factor of 5 compared to pure PSPE, reaching the same level as PBMA, which had a coverage of only 8%.

3. Discussion

To obtain amphiphilic zwitterionic copolymer coatings, we copolymerized SPE together with increasing amounts of hydrophobic BMA, while the content of photo-crosslinker BPMA

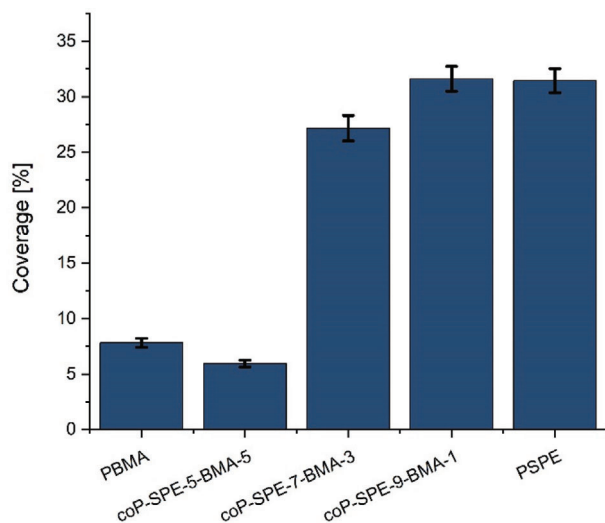


Figure 8. Fluorescence coverage after a 7-day dynamic field immersion on a rotating disk (shear stress 0.14 Pa). Coverages were determined on 3 slides with 60 fields of view each, error bars indicate the standard error ($n = 180$). All surfaces can be discriminated at a level of significance of 5% except PBMA to coP-SPE-5-BMA-5 and coP-SPE-9-BMA-1 to PSPE (post-hoc Tukey test).

was kept constant. Before the biological assays, the surface properties of the novel coatings were characterized by contact angle goniometry (WCA and CBCA) and spectroscopic ellipsometry. It was verified that the coatings were stable in saltwater for at least 7 days, corresponding to the duration of the longest assays, that is, the field exposures. The hydrophilicity of the coatings increased with increasing SPE content, both for the WCA measurements in air as well as for the CBCA analysis in saltwater. PBMA surfaces were considerably more hydrophobic than the other examined coatings (WCA = 73°) throughout the whole 7-day immersion period, even though the CBCA decreased over time to 42°. As expected, the purely hydrophilic PSPE coating did not show any marked change in contact angle during immersion (WCA of 25° and CBCA of 10°). The incorporation of hydrophobic BMA units into the SPE-based polymers shifted the WCA to more hydrophobic values (WCAs of 60° to 85°). Upon immersion, the amphiphilic coatings were prone to conformational rearrangement and contact angles between 10° and 14° were reached, which were very close to the values of the pure SPE-based coatings. Accordingly, we speculate that the hydrophilic sulfobetaine side chains of SPE became preferentially exposed at the surface, while the hydrophobic butyl side chains of BMA became buried inside the polymer films. The comparison of the changes between WCA and CBCA showed that the strongest rearrangement occurred in coatings of coP-SPE-7-BMA-3, which changed the CA by 72°. Both, coP-SPE-5-BMA-5 and coP-SPE-9-BMA-1 underwent smaller, but still substantial changes (56° and 50°, respectively). Thus, the amphiphilic coatings with 30% of hydrophobic BMA content showed the largest rearrangement underwater. For an amphiphilic coating of the carboxybetaine analogue to SPE copolymerized with 70 mol% of BMA, a similarly pronounced surface rearrangement upon immersion into aqueous media was recently reported. However,

comparisons to copolymers with different BMA contents were not undertaken.^[13] In the long term, all amphiphilic coatings showed CBCAs close to the one of pure PSPE. This suggests that extensive segmental rearrangement was possible even for the rather high BMA contents so that only the hydrophilic side chains of SPE were exposed at the interface.

The different extents of rearrangement of the polymer coatings were not only observable in the WCA and CBCA measurements, but also in the swelling kinetics in a saturated H₂O atmosphere. As expected, the hydrophobic PBMA films showed no noticeable swelling (0.1%). PSPE films in turn showed a strong thickness increase (34%), which is in good agreement with earlier studies.^[33] The amphiphilic copolymers showed intermediate swelling, with values of 24% for coP-SPE-9-BMA-1 ($k_w = 0.9 \text{ min}^{-1}$), 27% for coP-SPE-7-BMA-3 ($k_w = 1.6 \text{ min}^{-1}$), and 13% for coP-SPE-5-BMA-5 ($k_w = 1.4 \text{ min}^{-1}$). The initial increase of the layer thickness of the polymers due to water uptake was slightly higher than observed for dendritic polyglycerols in previous studies.^[44] However, with increasing time the diffusion limitation had a strong influence on the swelling curves, which is clearly visible when comparing the plots with non-diffusion limited models ($p = 1$). On the other hand, the fitted value of $p = 0.6$ indicated that the swelling was not purely diffusion limited either, as $p = 0.5$ would have been expected. Because of the constant cross-linker content of 1 mol% in all copolymers, an effect of differing degrees of crosslinking on the rearrangement dynamics could be excluded. Therefore, the observed differences in the rearrangement dynamics could be attributed to the different compositions of the copolymers, and to the number of side chains that needed to rearrange to reach an energetically favored conformation. Looking at the relative increase in thickness, coatings of coP-SPE-5-BMA-5 grew much less than coP-SPE-7-BMA-3 and coP-SPE-9-BMA-1. It is interesting to note that copolymer coP-SPE-7-BMA-3 showed a stronger swelling compared to coP-SPE-5-BMA-5 and reached even slightly higher thickness increases as compared to coP-SPE-9-BMA-1. At the same time, it showed the highest change in wettability upon immersion in water, which is also reflected in the fastest rearrangement kinetics.

Besides the wettability and the level of hydration, previous experiments showed that coatings with exclusively hydrophilic constituents are very soft and frequently prone to the incorporation of silt or other particulate matter present in marine environments.^[17] Laboratory sediment adsorption assays revealed decreasing levels of sediment incorporation into the copolymer coatings with increasing hydrophobic content, and thus with the reduced extent of swelling. Seemingly, a certain hydrophilicity, but also enough space within the polymer network is required to enable silt particles to stick to the antifouling coatings. An increase in the crosslinking density has proven to successfully reduce the sediment uptake as well. The incorporation of 50% BMA reduced the uptake of silt to one third compared to pure PSPE which was an effect similar to the increase of the crosslinker content from 1% to 2%.^[33]

The variation of the zwitterionic content significantly affected the anti-fouling properties of the coatings. The non-specific adsorption of fibrinogen decreased for all copolymers compared not only to films of pure PBMA. Above a PSPE content > 50 mol%, the adsorbed layer of protein was undetectably thin. The extremely low adsorption of fibrinogen on the amphiphilic

polymers is in line with previous reports.^[32,52] Also, the adhesion of lysozyme was significantly lower on all amphiphilic copolymers than on the pure PBMA and PSPE coatings.

The settlement of the marine diatom *N. perminuta* decreased with increasing hydrophilicity of the coatings. Pure PSPE films had the lowest settlement density. Yet, in experiments with zoospores of the green algae *U. linza*, the amphiphilic coatings outperformed the ones of pure PSPE and PBMA polymer coatings. Similar observations were made previously for amphiphilic PEG block copolymers.^[20,32,52] The good performance of amphiphilic zwitterionic polymers is in line with many results in literature showing high resistance against cell adhesion,^[30] attachment of *N. incertia*,^[29] and a good anti-fouling performance against zoospores of *U. linza*.^[53]

In recent studies, the formation of domains within amphiphilic coatings was described and it was pointed out that it could be energetically favorable if hydrophilic and hydrophobic islands are formed.^[21] In our study, no indications were found for such phase segregation, but future work to precisely understand the interplay of chemical composition and domain size would be very interesting.

In preliminary field tests, the highest fouling was accumulated on coatings with high SPE content (cf. Figure 8). This trend coincides well with the tendency of the coatings to accumulate silt (cf. Figure 7). Accumulation of biomass in the field exposures decreased with increasing hydrophobic content. Sample coP-SPE-5-BMA-5 performed similar to the hydrophobic control PBMA, and it is not clear where would be the optimum BMA content. It may be speculated whether coP-SPE-BMA copolymers with even lower levels of SPE incorporation (i.e., copolymers with $x < y$ in Figure 1) might further improve the anti-fouling effect in the field. The comonomer limited swelling due to the reduced hydrophilicity of such zwitterionic coatings, for instance, might enhance their underwater mechanical strength. Previous studies have shown that very soft and highly swollen hydrogels are prone to the accumulation of silt on which biomass can settle.^[17] Also, low amounts of carboxybetaines in a DCPEA matrix were sufficient to strongly improve the marine antifouling properties of the purely hydrophobic parent coatings.^[23] Similar observations were made for different hydrophilic functionalities, including zwitterionic ones, in a BMA-dominated polymer matrix.^[32]

4. Conclusion

A series of photo-crosslinked amphiphilic coatings from copolymers of the zwitterionic sulfobetaine methacrylate SPE with contents of hydrophobic butyl methacrylate BMA up to 50 mol% were prepared and characterized in detail. In comparison to not only the hydrophobic coating of homopolymer PBMA, which is known to show substantial fouling, but also to the hydrophilic zwitterionic coating of the homopolymer PSPE, which is thought to produce low-fouling surfaces, the incorporation of the hydrophobic comonomer enhanced the fouling resistance notably. This contrasts with the widespread perception that the incorporation of additional hydrophobic groups in strongly hydrophilic polymers and hydrogels is a priori detrimental to their high fouling resistance. The amphiphilic copolymers were equally effective against both proteins and fouling organisms in laboratory assays, as well as in short-term dynamic field tests in the ocean.

Upon exposure to a water saturated atmosphere, all polyzwitterion films swelled markedly, and the surfaces rearranged to form hydrogel coatings with a mostly hydrophilic surface, in contrast to the reference of pure PBMA. The extent of rearrangement and its kinetics changed slightly for different SPE:BMA ratios. The presence of hydrophobic BMA units in the coatings significantly improved the anti-fouling properties, resulting in lower protein adsorption, and lower settlement of *U. linza* zoospores than on coatings from both parent polymers. Further, the attachment of diatom *N. perminuta* was decreased more effectively. The incorporation of increasing amounts of hydrophobic BMA within the series also decreased the sediment uptake, which hypothetically is a consequence of the reduced swelling of such coatings, thus increasing their hardness in aqueous environments. When the amphiphilic coatings were challenged by a complex fouling community in the field, the amphiphilic copolymers showed substantially reduced biofouling compared to the zwitterionic homopolymer PSPE. Presumably, the lower sensitivity of the copolymers to silt uptake contributes crucially to the improved antifouling performance in the field. Notwithstanding this important effect, the incorporation of a certain amount of moderately hydrophobic elements into polyzwitterion coatings seems in general to boost the fouling resistance.

5. Experimental Section

Polymer Synthesis: Copolymers of 3-[N-2'-(methacryloyloxy)ethyl-N,N-dimethyl]ammonio propane-1-sulfonate (SPE), *n*-butyl methacrylate (BMA), and 2-(4'-benzoylphenoxy)ethyl methacrylate (BPMEA) were synthesized as described previously.^[9,38,54] A standard procedure^[38] of free radical copolymerization in solution was used employing the comonomer ratios specified in Table 1. Details are provided in the Supporting Information, including the ¹H NMR spectra (Figures S1–S3, Supporting Information). The polymers were purified by dialysis (membrane's nominal molecular weight cut-off MWCO = 3500 g mol⁻¹).

Chemicals and Sample Preparation: Glass substrates were modified with a monolayer of 3-aminopropyltrimethoxysilane (APTMS, 97%, Sigma Aldrich) to promote the adhesion of the polymer coatings on the surface following a previously described procedure.^[9,55] The substrates were treated in oxygen plasma for 3 min (0.4 mbar, 80 W, 22 kHz, miniFlecto-PC-MFC, Gala Instruments GmbH) and immersed into a 5% APTMS solution (v/v) under inert atmosphere (nitrogen, 99%, Air Liquide). Afterward, the substrates were immersed successively for 20 s into ethyl acetate (>99.8%, Fisher Scientific), ethanol (p.a., >99.8%, Fisher Scientific), and Milli-Q (Merck Millipore, Darmstadt, Germany) and ultrasonicated. Gold-coated glass substrates with a thin interlayer of titanium to promote adhesion were functionalized by a self-assembled monolayer of 11-amino-1-undecanethiol (ProChimia) following published protocols including a 24 h immersion and subsequent cleaning steps to remove non-covalently bound molecules.^[9] The polymers were dissolved in TFE (Roth) and spin-coated on the modified substrates (WS-650MZ-23NPP/Lite, Laurell Technologies Corporation) by following previously established protocols.^[9] The concentration of the polymers in the solutions was generally 1 wt% (0.5 wt% for SPR) to achieve thicknesses of the final coatings of 150 ± 8 nm. For SPR studies, polymer concentrations of the coating solutions were 0.5 wt%, yielding thicknesses of 55 ± 4 nm. The polymer coatings were covalently crosslinked by curing under UV-light (Uvacube 100, Dr. Hönle AG, 100 W iron-doped mercury vapor lamp, H1 Filter >310 nm, intensity at sample 15 mWcm⁻²) for 30 min.

Static Water Contact Angle (WCA) Goniometry: Static WCAs of the polymer coatings were determined using a custom-built goniometer equipped with a CCD camera. A droplet of triply distilled water was applied on the coating and its shape was fitted using Young's equation.

Captive Bubble Contact Angle (CBCA) Goniometry: Under water contact angle measurements were performed using a custom-build setup^[56] with a CCD camera and a glass cell filled with saltwater as described by Kester.^[42] A slide with the coated side facing down was placed in the cell, and a 20 µL air bubble was deposited using a U-shaped syringe from underneath the coated surface at ambient temperature. The water/solid contact angle at the substrate–air–water interface was fitted manually.^[56] The reported values were averages of five measurements. To monitor the rearrangement of the polymer surfaces, contact angles were measured after 1, 5, 10, 15, 30, 45, 60, and 120 min and after 1, 2, 4, and 7 days of immersion.

Spectroscopic Ellipsometry: The thickness of dry polymer films was measured by spectroscopic ellipsometry (M-2000, Woollam, USA with the CompleteEase software package) at incidence angles of 65°, 70°, and 75°. A single layer of polymer was modeled using an absorbing film model. The water uptake from a saturated vapor atmosphere in a humidity chamber was also followed by spectroscopic ellipsometry. As H₂O was present with a large excess compared to the volume of the coating, the water concentration was assumed to be constant to apply pseudo first-order kinetics given by Equation (1),

$$\frac{d(t)}{d_e} = 1 - e^{-k_w(t-t_0)^p} \quad (1)$$

where k_w represents the rate of water uptake, t_0 the inset time of the swelling, $d(t)$ the time dependent thickness at time t and d_e the final thickness in equilibrium with water. Beyond a normal Langmuir kinetics, the parameter p was used to account for a diffusion limited water uptake of the part of the polymer film closer to the substrate.^[45] While for very thin films p should converge to 1 (Langmuir case), thicker films face an additional time delay due to the diffusion of water leading to values <1 . To account for this diffusion, a formalism recently developed was applied to describe adsorption processes from very diluted solutions using a fixed exponent of 0.5.^[45] To account for intermediate behavior, the exponent was fitted globally for the measured samples between the extremes of 0.5 and 1.

Stability Test: The stability of the polymer coatings in saltwater was monitored by the dry film thickness after 1 and 2 h and 1, 2, 3, and 7 days of immersion in freshly prepared seawater.^[42,57] Before the measurement the coatings were rinsed with Milli-Q water to remove the remaining salt from the surface.

Surface Plasmon Resonance (SPR) Spectroscopy: The protein adsorption on the different coatings was analyzed by SPR (SRC7000DC, pump SR7500, autosampler SR7100, Reichert). After the injection of PBS buffer with a flow rate of 200 µL·min⁻¹ until a stable baseline was obtained, the flow rate was reduced to 10 µL·min⁻¹. Protein solutions (1 mg·mL⁻¹, in PBS) were injected for 10 min at a flow rate of 10 µL·min⁻¹. Washing with PBS for another 15 min at 10 µL·min⁻¹ removed reversibly attached proteins. The difference in the shift of the SPR signal before and after protein adsorption correlated with the amount of protein irreversibly attached to the surface.

Dynamic Attachment Assays of *Navicula perminuta*: Dynamic microfluidic measurements of the accumulation of diatoms followed previously published protocols.^[48,58] A suspension with a concentration of 3.0×10^6 cells·mL⁻¹ in filtered seawater (FSW, CaribSea, USA, pH 8.0) was prepared from a continuous *N. perminuta* culture. IBIDI sticky slides 0.1 (IBIDI) were placed onto the substrates coated with the polymers to form a microfluidic channel (150 µm height). The suspension was pumped through the channel for 90 min exerting a constant wall shear stress of 0.16 Pa followed by a 10 min rinsing step with FSW to remove non-settled diatoms. For quantification, 30 fields of view from the central part of the channel were acquired by a phase-contrast microscope (Nikon Eclipse TE2000-U, 10× phase contrast objective Nikon CFI Plan Fluor DLL NA 0.3). Automatic diatom counting was conducted using the NIS Elements software (Nikon). Statistical analysis was performed by ANOVA with post-hoc Tukey test at a significance level of $\alpha = 0.05$.

Settlement and Removal of Zoospores of *Ulva linza*: Zoospores were collected from mature plants of *U. linza* from Craster, U.K. Following the method of Callow et al.,^[50] a zoospore suspension (10 mL; 1×10^6 spores

mL⁻¹) was added to the compartments of quadriPERM dishes (Sarstedt Ltd.) containing 6 replicates of each sample. After 45 min in the dark at 20 °C, the slides were passed backward and forward 10 times through a beaker of 0.22 µm filtered ASW (Tropic Marin salts) to wash off unsettled (i.e., swimming) spores. Slides were fixed with ASW containing 2.5% glutaraldehyde (Thermo Fisher Scientific, USA). The density of zoospores on the surface per slide was counted using an image analysis system attached to a fluorescence microscope (Zeiss Axioscope 2 plus). Spores were visualized and counted using the autofluorescence of chlorophyll (excitation and emission: 546 and 590 nm) on 30 fields of view of 0.15 mm² each per slide. Zoospore removal was determined using another three slides with zoospores settled for 45 min by the method above. Slides were exposed to a shear stress of 52 Pa in a specially designed turbulent flow water channel.^[59] The slides were then fixed and the number of spores remaining attached was counted. The counts were averaged over all replicates and statistical analysis was performed by ANOVA with post-hoc Tukey test at a significance level of $\alpha = 0.05$.

Photometric Measurement of Sediment Incorporation: Natural sand sediment from the test site at Port Canaveral (USA) was autoclaved and suspended in sterile filtered seawater (CaribSea Pure, Fort Pierce, FL, USA) at a concentration of 37.5 g·L⁻¹. The polymer-coated slides were placed horizontally on a linear shaker (Laboshake Heavy Load, Gerhardt Analytical Systems). After 2 h at 60 rpm, the slides were shaken twice vertically for 2 min in sediment-free saltwater to remove non-sticking particles. On each sample surface the photometric absorption at 440 nm was measured and compared to the absorbance of the pristine slide.

Dynamic Short-Term Field Exposure on a Rotating Disk: Field testing was executed on a custom-built rotating disk at Port Canaveral, Florida, USA (28°24'29.4"N 80°37'37.7"W) following previously published protocols.^[60] The calculated shear force at the center of the samples was 0.14 Pa. After a 7-days immersion, the absorbed material was treated with 2.5 wt% glutaraldehyde (Fisher Scientific, Pittsburg PA, USA) for fixation. The absorbed material was quantified using the fluorescence coverage of 60 fields of view (light source: ExFO XCite-120, Excelitas Technologies Corp., filter: BV-2A, excitation: 400–440 nm, dichroic mirror: 455, barrier filter: 470, Nikon Instruments).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

Research data are not shared.

Keywords

amphiphilic, field tests, fouling release, marine biofouling, *Navicula perminuta*, polyzwitterion, *Ulva linza*

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